

Toni V. Panzera

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Research Interests

Young stars, T Tauri stars, brown dwarfs, circumstellar disks, stellar activity, accretion physics

Education

Rice University, Houston TX

Ph.D., Astrophysics

Anticipated 2028

- Thesis Title: Accretion Properties of Very Low Mass Stars
- Thesis Advisor: Prof. Christopher M. Johns-Krull

M.S., Astrophysics

2024

University of Denver, Denver CO

B.S., Mathematics & Physics (Hons)

2022

- Thesis Title: A Spectropolarimetric Study of the Southern Hemisphere Wolf-Rayet Stars WR 12 and WR 71
- Thesis Advisor: Prof. Jennifer L. Hoffman

Publications

Panzera, T.V., Flagg, L.S., Mueller, M.A., Johns-Krull, C.M. & Herczeg, G.J. 2026 “Accretion Properties of the Young Brown Dwarf 2MASS J08440915-7833457,” *Astrophysical Journal*, 999, 111.

Talks & Posters

Panzera, T.V., “Probing Accretion at the Lowest Masses: The Young Brown Dwarf J0844,” contributed talk at Southwest Star & Planet Formation Conference, December 2025

Panzera, T.V. & Johns-Krull, C.M., “Bridging the Star-Planet Gap: Accretion Properties of the Young Brown Dwarf J0844,” poster presented at Role of Accretion And Ejection Variability in Young Stellar Objects (RAVEYSO) Conference, Garching, Germany, May 2025.

Panzera, T.V. & Johns-Krull, C.M., “Accretion Properties of the Young Brown Dwarf J0844,” poster presented at Cool Stars Conference, June 2024.

Panzera, T.V., “Accretion onto Very Low Mass Stars and Brown Dwarfs,” contributed talk at the Hubble Space Telescope (HST) UV Legacy Library of Young Stars as Essential Standards (ULLYSES) conference at Space Telescope Science Institute (STScI), March 2024.

Observing Experience

Harlan J. Smith Telescope, McDonald Observatory

- Exoplanet transits (Echelle spectrograph, 4 nights)
- IR spectra of YSOs (IGRINS, 4 nights)

December 2022

October 2025

Hale Telescope, Palomar Observatory

- IR spectra of brown dwarfs (TripleSpec, 2 nights)

January 2025

Skills

Computer Programming & Data Analysis

- Python, R, Mathematica, LabVIEW
- Git, GitHub
- SAOImage DS9
- RADMC-3D